

CS543 – Web Services

Assignment 4 – Part A: Model the CampusEats Orders Service

A1. Pick your service

Selected Service: Orders

The Orders service is responsible for managing carts, cart items, orders, order items, and order status. The service owns the order-related data and operations.

A2. List the operations we would have written as SOAP

- createOrder(...)
- getOrder(...)
- addItemToCart(...)
- updateOrderStatus(...)
- cancelOrder(...)

These SOAP-style operations are the starting point for converting the Orders service into REST resources.

A3. Find the nouns

SOAP Operation	REST Resource
createOrder(...)	/orders
getOrder(...)	/orders/{id}
addItemToCart(...)	/carts/{id}/items
updateOrderStatus(...)	/orders/{id}/status
cancelOrder(...)	/orders/{id}/cancellation

The HTTP method tells us what action is being performed:

- POST /orders → create an order
- GET /orders/42 → read an order
- POST /carts/10/items → add an item to a cart
- POST /orders/42/cancellation → request cancellation

A4. Resource Table

Method	URL	What it does	Success Code	Failure Codes
POST	/orders	Creates a new order	201 Created	400, 422
GET	/orders/{id}	Retrieves one order using its ID	200 OK	404
GET	/orders?status=CONFIRMED	Returns orders filtered by status	200 OK	400
PUT	/orders/{id}/status	Updates the status of an order	200 OK	404, 409, 422
POST	/orders/{id}/cancellation	Requests cancellation of an order	200 OK	404, 409, 422

A5. Justification of the difficult choice

The most difficult operation to map to REST is `cancelOrder(...)` because cancellation is a state-changing operation that can only be performed when the order is in an appropriate state. Instead of using a verb-based URL such as `/orders/42/cancelOrder`, I represented the cancellation as a sub-resource using `/orders/42/cancellation`. I rejected the verb-based URL because REST URLs should represent resources rather than operations, while the HTTP method communicates the action.